



BIOLOGY CH.8 — SENSORY ORGANS — FULL MIND MAP

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Topics in this chapter:

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7. Sense of Smell (Olfaction)
8. Anaesthetics & Sensation Loss
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1. Types of Sensory Receptors

- Gustatory receptors tongue par hote hain - taste sense ke liye. [Q510, Q531]
- Olfactory receptors nose mein hote hain - smell/odour molecules detect karte hain. [Q514, Q531]
- Thermoreceptors nerve endings mein temperature changes detect karte hain. [Q510]
- Phonoreceptors inner ear mein hote hain - hearing sense ke liye. [Q510]
- Photoreceptors eyes mein hote hain - light detect karte hain. [Q510]
- Sensory receptors overall eyes, ears, nose, mouth, internal organs mein distributed hote hain. [Q531]

2. Human Eye - Structure & Function

- Cornea aur Retina eye mein paaye jate hain. Eye ke parts: Pupil (light admits), Iris (coloured tissue), Retina (images → electrical impulses), Sclera (outermost coat), Cornea (clear anterior portion, 2/3 optical power). [Q511, Q521]
- Iris eye ka coloured part hai jo pupil ke aaspaas hai - light entry ko control karta hai pupil size adjust karke. [Q526]
- Pupil iris mein hole hai jisse light pass hoti hai. [Q526]
- Cornea clear anterior coat hai jo 2/3 optical power provide karta hai. [Q526]
- Retina sensitive layer hai jo images ko electrical impulses mein convert karti hai. [Q526]
- Fovea retina ka chhota depression hai jahan visual acuity (drishti teevrata) sabse zyada hoti hai. [Q520]
- Blind spot optic disc ki position ko correspond karta hai jahan koi photoreceptor nahi hote. [Q520]
- Vitreous body lens aur retina ke beech clear gel hai. [Q520]
- Choroid sclera aur retina ke beech ki middle layer hai. [Q520, Q521]
- Ciliary body ciliary muscle contain karti hai jo lens shape change karti hai near focusing ke liye. [Q521]
- Optic disc pe ganglion cell axons eye se bahar nikalte hain → optic nerve banti hai. [Q517]

3. Human Eye - Photoreceptors (Rods & Cones)

- Rods aur Cones dono retina mein paye jate hain - ye photoreceptor cells hain. [Q517, Q524, Q529]
- Cones (shanku): bright/normal light ke liye sensitive hain, colour sensation dete hain (3 colours: blue, green, red), photopic vision. [Q517, Q524, Q529]
- Rods: dim/low light ke liye sensitive hain, twilight/scotopic vision dete hain, colour nahi dekh paate. [Q517, Q524, Q529]

4. Human Ear - Structure & Function

- Ear 3 parts mein divided hai: - Outer ear: Pinna + External auditory meatus (sound waves collect/transmit) - Middle ear: Malleus (Hammer), Incus (Anvil), Stapes (Stirrup) - 3 ossicles - Inner ear: Cochlea, Semicircular canals, Vestibule [Q513, Q525, Q528]
- Tympanic membrane (eardrum) ear canal ke end mein hai - outer ear ko middle ear se separate karti hai, sound vibrations ke according vibrate karti hai. [Q513, Q525]
- Middle ear ke 3 bones (ossicles) sound ko amplify karte hain - Malleus/Hammer (eardrum se attached), Incus/Anvil, Stapes/Stirrup (membrane se attached). Sound 20 times tak amplify hoti hai. [Q516, Q528, Q530]
- Cochlea inner ear mein fluid-filled, spiral-shaped cavity hai - pressure variations ko electrical signals mein convert karti hai (auditory transduction). [Q522, Q530]
- Auditory canal pinna se eardrum tak sound transmit karti hai. Pinna sound waves collect karta hai. [Q522]
- Eustachian tubes middle ears ko throat se connect karte hain - fluid drain karte hain aur air pressure equalize karte hain. [Q525]

5. Human Ear - Balance

- Ear sound ke alawa balance maintain karne mein bhi help karta hai. [Q515]
- Vestibular system ear ke andar hai: 3 semicircular canals + 2 otolith organs (Utricle aur Saccule). Har semicircular canal specific head movement direction ke liye responsible hai. [Q515]

6. Sense of Taste (Gustation)

- Gustatory receptors tongue par taste buds ke roop mein hote hain. [Q510, Q531]
- 5 basic tastes: Salty, Sour (ion channels se), Sweet, Bitter, Umami (G protein-coupled receptors se). [Q531]
- Tongue ek unique muscle hai jo dono ends par attached NAHI hai - independently function karti hai: tasting, gulping, digestive enzymes provide karna. [Q512]
- Largest muscle: Gluteus maximus. Smallest muscle: Stapedius. [Q512]

7. Sense of Smell (Olfaction)

- Olfactory receptors nose mein hote hain - protein odour molecules ko bind karte hain. [Q514]
- Olfactory indicators smell se kaam karte hain - acid/base ke saath smell change hoti hai. Examples: Vanilla, Onion, Clove. [Q523]
- Visually impaired students olfactory indicators use kar sakte hain. Visual indicators: Turmeric, grape juice, petunia leaves, litmus. [Q523]

8. Anaesthetics & Sensation Loss

- Anaesthetic agent local ya general loss of sensation produce karta hai - temporarily nerves se brain tak sensory signals block karta hai. [Q518]
- Anaesthetics surgical operations mein use hote hain areas numb karne ya sleep induce karne ke liye. Examples: Desflurane, Isoflurane, Nitrous oxide, Sevoflurane, Xenon. [Q518]

9. Other Sensory Facts

- Organ of Corti hearing se related hai (auditory transduction) - inner ear mein hai, 3 rows outer hair cells + 1 row inner hair cells. Sound waves → eardrum → vibrations bend stereocilia on hair cells. [Q519]
- Auditory system = hearing. Visual sensory system = colour, light, contrast, motion. [Q514]
- Cartilage connective tissue hai jo ears mein hoti hai jisse ears fold kar sakte hain - ye bone surfaces ko joints par smooth bhi karti hai, nose/trachea/larynx mein bhi hoti hai. [Q527]
- Connective tissues ke types: Areolar, Adipose, Bone, Tendon, Ligament, Blood. [Q527]
- Cochlea pressure variations ko electrical signals mein convert karti hai auditory nerve ke through brain tak bhejne ke liye. [Q528]